

Particle Swarm Optimization: A Novel Approach

Step 1: Edge Prediction

The predicted congestion on edge (i, j) is

$$\hat{y}_{ij} = \text{MLP}(h_i^{\text{final}} \parallel h_j^{\text{final}} \parallel f_{ij})$$

where h_i^{final} is the node embedding vector of node i and h_j^{final} is the node embedding vector of node j . f_{ij} denotes the forecasted time and weather parameters on edge ij .

Denormalization

$$y_{ij} = \hat{y}_{ij}\sigma_y + \mu_y$$

where $y_{ij} = q_{ij}$ is the predicted traffic flow (veh/hr).

Step 2: Congestion Threshold Computation

Each edge has a user-specified road capacity C_i entered by the traffic engineer. Instead of using arbitrary weighting factors, congestion is evaluated using the critical volume-to-capacity ratio, a standard traffic engineering concept.

The congestion threshold is

$$T_i = \eta_i C_i$$

where:

- C_i : road capacity (veh/hr)
- η_i : critical utilization factor.

Typical values are

$$\eta_i = \begin{cases} 0.85, & \text{Urban arterial} \\ 0.90 - 0.95, & \text{Urban expressway} \end{cases}$$

For the Mumbai case study, $\eta_i = 0.85$ for arterials and $\eta_i = 0.95$ for expressways (assumption: the traffic congestion begins at LOS D). An edge is classified as congested whenever $q_i > T_i$.

(Source: Civil Department, IIT Bombay)

Table 1: The LOS of a Mid Block Section

LOS	Quality	Speed (kmph)	V/C	Description
A	Free-flow	80	0.6	High level of physical and psychological comfort
B	Reasonable free-flow	70	0.7	Reasonable level of physical and psychological comfort
C	Near free-flow	60	0.8	Local deterioration possible with blockages
D	Medium flow	50	0.85	Non-recoverable local disruptions
E	At capacity flow	40	0.9	Minor disturbances resulting breakdown
F	Congested flow	15	1.0	Break down of flow capacity drops

Step 3: Excess Congestion

The amount of traffic exceeding the acceptable operating level is

$$E_i = \max(0, q_i - T_i)$$

Only congested edges participate in redistribution.

Step 4: Particle Swarm Optimization

Each particle represents one complete traffic signal timing plan:

$$x_p = [C_A, C_B, C_C, C_D, C_E]$$

where C_i denotes the cycle time at junction i . The current implementation uses **30 particles** over **100 iterations**.

Velocity Update

$$v_p^{k+1} = wv_p^k + c_1r_1(p_{\text{best}} - x_p^k) + c_2r_2(g_{\text{best}} - x_p^k)$$

where, $w = 0.7$, $c_1 = c_2 = 1.5$ (assumed)

Position Update

$$x_p^{k+1} = x_p^k + v_p^{k+1}$$

Step 5: Signal Timing

Step 5.1: Equations for the Dynamic Green Split Ratio

For any edge i entering intersection node j :

A. Flow Ratio (y_i)

The demand-to-capacity relationship of edge i is captured by the flow ratio $y_i = \frac{q_i}{s_i}$

Where:

q_i = Actual/predicted traffic flow rate on approach i (veh/hr).

s_i = Saturation flow rate (veh/hr), calculated as

$$s_i = \text{base_flow} \times \text{lanes}_i$$

base_flow = 2200 veh/hr for Expressway segments.

base_flow = 1900 veh/hr for Arterial segments.

B. Dynamic Green Split Allocation (λ_i)

To distribute green time proportionally while respecting the intersection's capacity bounds, we use the following allocation:

$$\text{If } \sum_{k \in \text{incoming}(j)} y_k > 0 : \lambda_i = S_j \times \frac{y_i}{\sum_{k \in \text{incoming}(j)} y_k}$$

$$\text{If } \sum_{k \in \text{incoming}(j)} y_k = 0 \quad (\text{no traffic}): \lambda_i = \text{base_split}_i$$

Where S_j is the total baseline split budget available at intersection j (sum of baseline splits of all approaches entering j). base_split_i is the hardcoded default split for that approach.

C. Physical Clearance Safety Cap

To ensure there is sufficient time reserved for yellow clearance and pedestrian crossings, the split is capped at a maximum of 90% of the cycle: $\lambda_i^{\text{final}} = \min(0.90, \lambda_i)$

D. Green Time in Seconds (g_i)

Finally, the actual green duration in seconds displayed on the frontend is: $g_i = \lambda_i^{\text{final}} \times C_j$ where C_j is the optimized cycle time of node j in seconds.

Underlying Theorem: The Equisaturation Principle

The dynamic green split equation is derived from the Equisaturation Principle (also known as the Equal Degree of Saturation Principle), which is the cornerstone of Webster's Method for traffic signal optimization, which states:

Average delay for vehicles at a signalized intersection is minimized when the degree of saturation (x) is equal across all conflicting phases. $x_1 = x_2 = \dots = x_n = X$

Mathematical Derivation

1. **Definition of Degree of Saturation (x_i):** The degree of saturation is the ratio of approach volume (q_i) to its capacity. The capacity of edge i is $s_i \times \lambda_i$, so:

$$x_i = \frac{q_i}{s_i \times \lambda_i} = \frac{y_i}{\lambda_i}$$

2. **Equating Degrees of Saturation:** According to the principle, we set $x_i = X$ for

all approaches: $\frac{y_i}{\lambda_i} = X \implies \lambda_i = \frac{y_i}{X}$

3. **Summing the Splits:** The sum of all splits at an intersection must equal the total

split budget S : $\sum_i \lambda_i = S \implies \sum_i \frac{y_i}{X} = S$ Solving for X : $X = \frac{\sum_i y_i}{S}$

4. **Substituting X Back into the Split Equation:** $\lambda_i = \frac{y_i}{X} = \frac{y_i}{\frac{\sum_k y_k}{S}} = S \times \frac{y_i}{\sum_k y_k}$

This completes the proof, showing that allocating green splits proportionally to the flow ratios (y_i) mathematically balances intersection load and minimizes cumulative queue delay.

The effective green time allocated to an approach is

$$g_i = \lambda_i C_i$$

where C_i is the optimized cycle time and λ_i is the green split ratio. Discharge capacity is computed using the Highway Capacity Manual relationship:

$$D_i = s_i \left(\frac{g_i}{C_i} \right)$$

where:

- D_i : discharge capacity (veh/hr)
- s_i : saturation flow (veh/hr)

Since $g_i = \lambda_i C_i$, the discharge simplifies to

$D_i = s_i \lambda_i$ which is the effective service capacity of the approach.

Step 6: Transferable Flow

The transferable traffic from edge i is constrained by excess congestion and discharge capacity. Therefore,

$$F_i = \min(E_i, D_i)$$

No heuristic percentages are introduced.

Step 7: Downstream Receiving Capacity

For downstream edge k , the remaining acceptable capacity is

$$R_k = \max(0, T_k - q_k)$$

If $R_k = 0$, the downstream edge cannot receive additional vehicles.

Step 8: Flow Allocation

Suppose multiple upstream edges feed edge k . If $\sum_i F_i \leq R_k$, all flows are accepted. Otherwise, incoming flows are proportionally scaled:

$$\hat{F}_i = F_i \frac{R_k}{\sum_j F_j}$$

This guarantees $\sum_i \hat{F}_i = R_k$.

Step 9: Vehicle Conservation

The upstream congestion becomes

$$q_i^{\text{new}} = q_i - \hat{F}_i$$

The downstream congestion becomes

$$q_k^{\text{new}} = q_k + \sum_i \hat{F}_i$$

Thus,

$$\sum q^{\text{new}} = \sum q, \text{ ensuring conservation of vehicles across the network.}$$

Step 10: PSO Objective Function

The Particle Swarm Optimization minimizes the network-wide congestion index:

$$\sum_{i=1}^N \left(\frac{q_i}{T_i} \right)^2$$

subject to:

- $30 \leq C_i \leq 120$
- $0 < \lambda_i < 1$
- $q_i \geq 0$
- $R_k \geq 0$

The squared penalty strongly discourages congestion levels above the threshold while allowing the optimizer to redistribute traffic over less congested routes.

Scenario 1 — Early Morning (3:00–5:30 AM)

Novel PSO Traffic Optimization

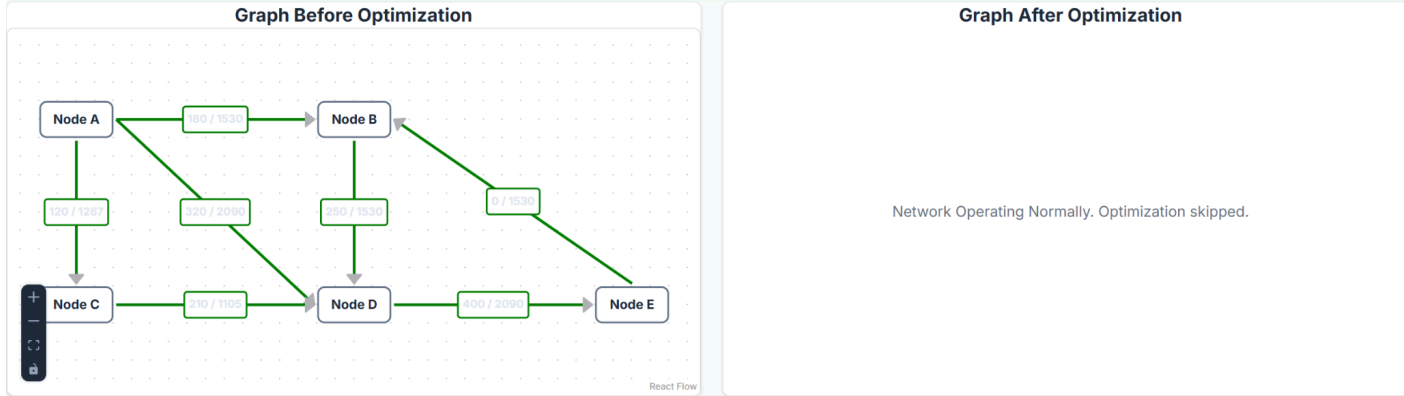
Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	180
A→C	1490	120
A→D	2200	320
B→D	1800	250
C→D	1300	210
D→E	2200	400
E→B	1800	0

Run Optimization

Optimization Status: No Optimization Required. All edges are operating below their capacity thresholds.



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	180	1530	180	54s / 54s	Normal	Normal
A→C	1490	50	3	0.95	Arterial	120	1267	120	22.5s / 22.5s	Normal	Normal
A→D	2200	60	4	1.6	Expressway	320	2090	320	28.18s / 28.18s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	250	1530	250	33.99s / 33.99s	Normal	Normal
C→D	1300	40	2	0.8	Arterial	210	1105	210	42.83s / 42.83s	Normal	Normal
D→E	2200	60	4	1.4	Expressway	400	2090	400	22.5s / 22.5s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	0	—
Average congestion (veh/hr)	211.43	—
Maximum congestion (veh/hr)	400.00	—
Total network congestion	1480.00	—
Mean congestion-to-threshold ratio	0.13	—
Average signal cycle time (s)	—	—
Fitness value	0.00	0.00
PSO iterations to convergence	—	0

Scenario 2 — Morning Build-up (6:00–8:00 AM)

Novel PSO Traffic Optimization

Research Prototype

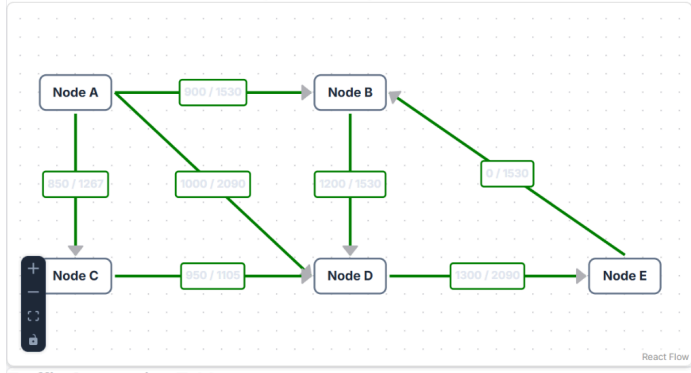
Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	900
A→C	1490	850
A→D	2200	1000
B→D	1800	1200
C→D	1300	950
D→E	2200	1300
E→B	1800	0

Run Optimization

Optimization Status: No Optimization Required. All edges are operating below their capacity thresholds.

Graph Before Optimization



Graph After Optimization

Network Operating Normally. Optimization skipped.

Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	900	1530	900	54s / 54s	Normal	Normal
A→C	1490	50	3	0.95	Arterial	850	1267	850	22.5s / 22.5s	Normal	Normal
A→D	2200	60	4	1.6	Expressway	1000	2090	1000	20.78s / 20.78s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	1200	1530	1200	38.5s / 38.5s	Normal	Normal
C→D	1300	40	2	0.8	Arterial	950	1105	950	45.72s / 45.72s	Normal	Normal
D→E	2200	60	4	1.4	Expressway	1300	2090	1300	22.5s / 22.5s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	0	—
Average congestion (veh/hr)	885.71	—
Maximum congestion (veh/hr)	1300.00	—
Total network congestion	6200.00	—
Mean congestion-to-threshold ratio	0.57	—
Average signal cycle time (s)	—	—
Fitness value	0.00	0.00
PSO iterations to convergence	—	0

Scenario 3 — Mumbai Morning Peak (8:30–10:30 AM)

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	1750
A→C	1490	1600
A→D	2200	1500
B→D	1800	2300
C→D	1300	1700
D→E	2200	1800
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED	ITERATIONS	INITIAL FITNESS	FINAL FITNESS
Yes	100	8.79	8.4

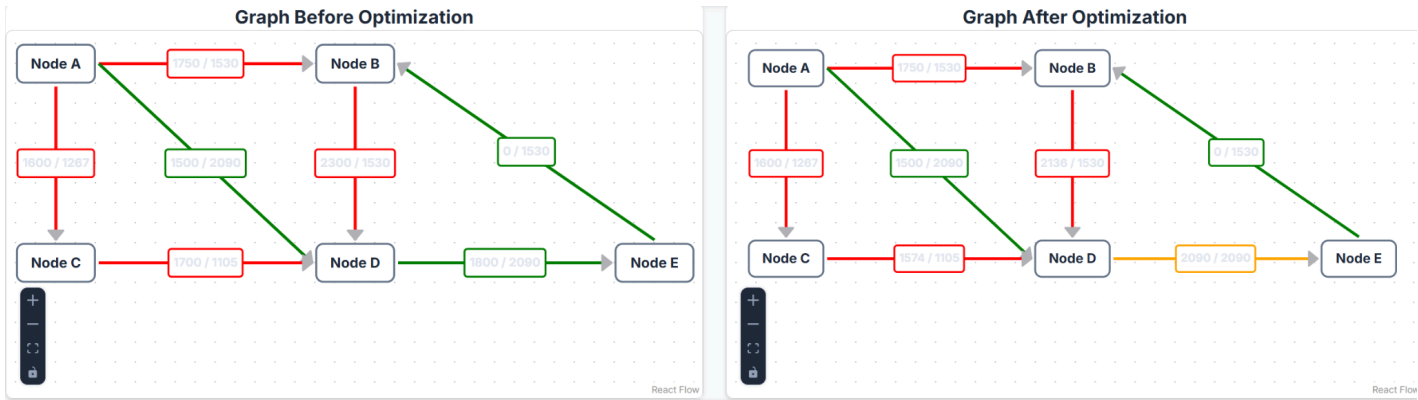
Overall Fitness Improvement: 4.44%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	31.45	-21.4%
B	60	67.21	12.0%
C	45	39.15	-13.0%
D	70	99.35	41.9%
E	50	36.88	-26.2%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	60.49s	+12.0%
A→C	22.5s	19.58s	-13.0%
A→D	17.52s	26.48s	+51.1%
B→D	41.48s	58.22s	+40.4%
C→D	45.99s	64.33s	+39.9%
D→E	22.5s	16.6s	-26.2%
E→B	0s	0s	0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	1750	1530	1750	54s / 60.49s	Congested	Congested
A→C	1490	50	3	0.95	Arterial	1600	1267	1600	22.5s / 19.58s	Congested	Congested
A→D	2200	60	4	1.6	Expressway	1500	2090	1500	17.52s / 26.48s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	2300	1530	2136	41.48s / 58.22s	Congested	Congested
C→D	1300	40	2	0.8	Arterial	1700	1105	1574	45.99s / 64.33s	Congested	Congested
D→E	2200	60	4	1.4	Expressway	1800	2090	2090	22.5s / 16.6s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	4	4
Average congestion (veh/hr)	1521.43	1521.43
Maximum congestion (veh/hr)	2300.00	2136.41
Total network congestion	10650.00	10650.00
Mean congestion-to-threshold ratio	1.00	0.99
Average signal cycle time (s)	—	54.81
Fitness value	8.79	8.40
PSO iterations to convergence	—	100

Scenario 4 — Midday (11:00 AM–3:00 PM)

Novel PSO Traffic Optimization

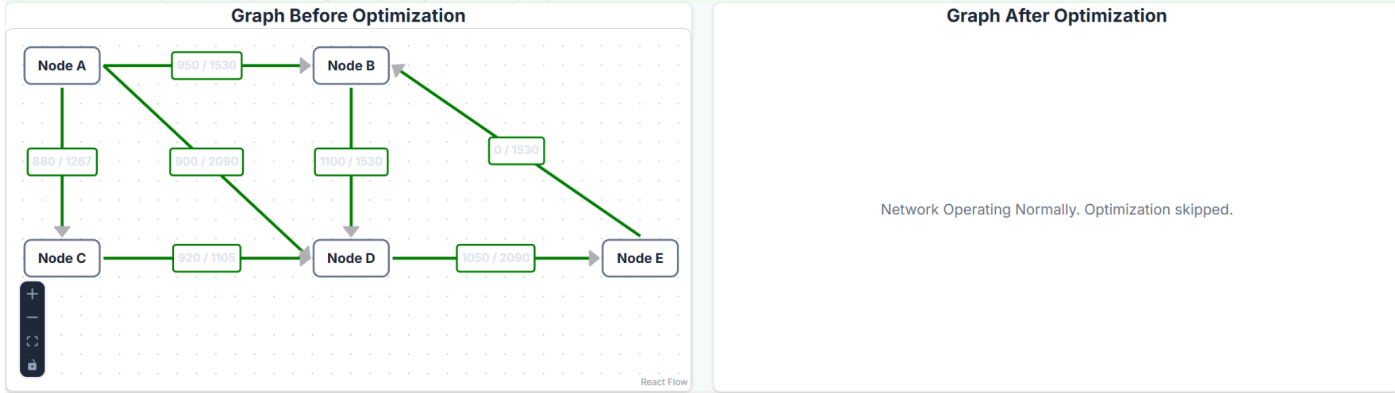
Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	950
A→C	1490	880
A→D	2200	900
B→D	1800	1100
C→D	1300	920
D→E	2200	1050
E→B	1800	0

Run Optimization

Optimization Status: No Optimization Required. All edges are operating below their capacity thresholds.



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	950	1530	950	54s / 54s	Normal	Normal
A→C	1490	50	3	0.95	Arterial	880	1267	880	22.5s / 22.5s	Normal	Normal
A→D	2200	60	4	1.6	Expressway	900	2090	900	19.98s / 19.98s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	1100	1530	1100	37.71s / 37.71s	Normal	Normal
C→D	1300	40	2	0.8	Arterial	920	1105	920	47.31s / 47.31s	Normal	Normal
D→E	2200	60	4	1.4	Expressway	1050	2090	1050	22.5s / 22.5s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	0	—
Average congestion (veh/hr)	828.57	—
Maximum congestion (veh/hr)	1100.00	—
Total network congestion	5800.00	—
Mean congestion-to-threshold ratio	0.54	—
Average signal cycle time (s)	—	—
Fitness value	0.00	0.00
PSO iterations to convergence	—	0

Scenario 5 — School Closing / Lunch Rush (3:00–5:00 PM)

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	1400
A→C	1490	1250
A→D	2200	1200
B→D	1800	1700
C→D	1300	1400
D→E	2200	1500
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED Yes	ITERATIONS 100	INITIAL FITNESS 5.5	FINAL FITNESS 5.03
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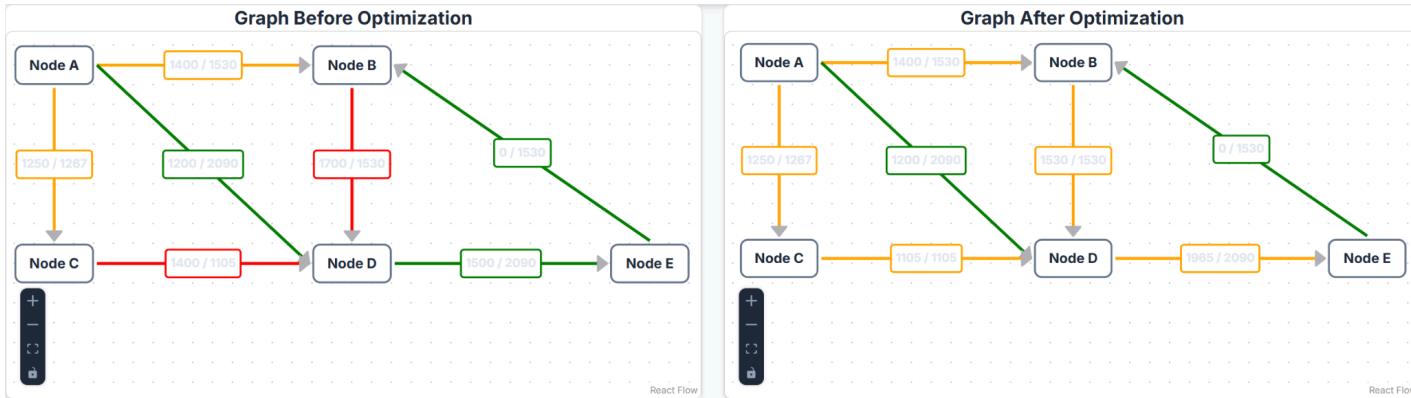
Overall Fitness Improvement: 8.55%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	93.81	134.5%
B	60	45.76	-23.7%
C	45	83.8	86.2%
D	70	89.82	28.3%
E	50	81.07	62.1%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	41.18s	-23.7%
A→C	22.5s	41.9s	+86.2%
A→D	17.83s	26.41s	+48.1%
B→D	39s	51.99s	+33.3%
C→D	48.17s	56.32s	+16.9%
D→E	22.5s	36.48s	+62.1%
E→B	0s	0s	0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	1400	1530	1400	54s / 41.18s	Normal	Normal
A→C	1490	50	3	0.95	Arterial	1250	1267	1250	22.5s / 41.9s	Normal	Normal
A→D	2200	60	4	1.6	Expressway	1200	2090	1200	17.83s / 26.41s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	1700	1530	1530	39s / 51.99s	Congested	Normal
C→D	1300	40	2	0.8	Arterial	1400	1105	1105	48.17s / 56.32s	Congested	Normal
D→E	2200	60	4	1.4	Expressway	1500	2090	1965	22.5s / 36.48s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	2	0
Average congestion (veh/hr)	1207.14	1207.14
Maximum congestion (veh/hr)	1700.00	1965.00
Total network congestion	8450.00	8450.00
Mean congestion-to-threshold ratio	0.80	0.77
Average signal cycle time (s)	—	78.85
Fitness value	5.50	5.03
PSO iterations to convergence	—	100

Scenario 6 — Evening Peak (5:30–8:30 PM)

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	1850
A→C	1490	1750
A→D	2200	1900
B→D	1800	2500
C→D	1300	1950
D→E	2200	2150
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED	ITERATIONS	INITIAL FITNESS	FINAL FITNESS
Yes	100	11.04	10.98

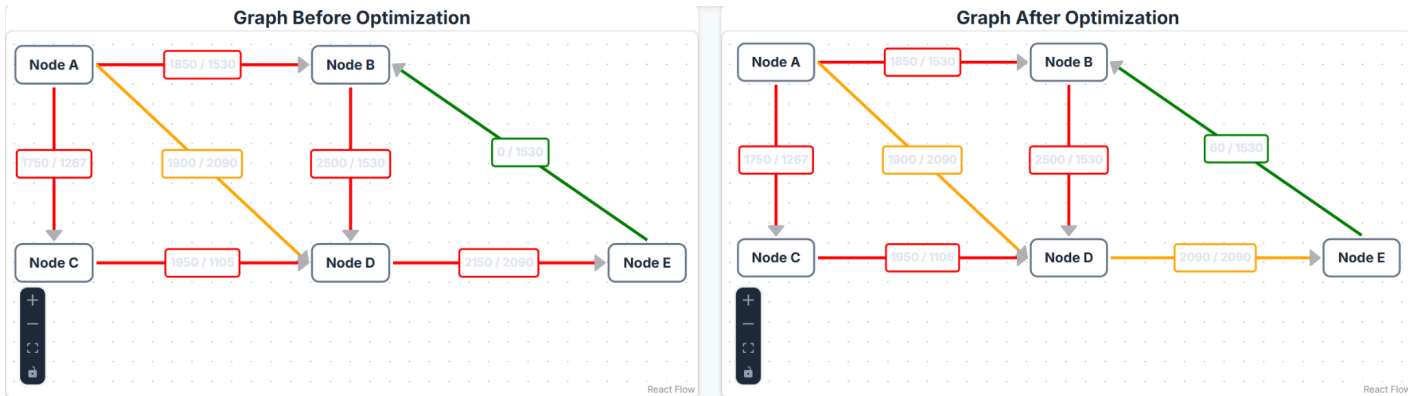
Overall Fitness Improvement: 0.54%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	59.66	49.1%
B	60	45.39	-24.3%
C	45	63.1	40.2%
D	70	84.56	20.8%
E	50	80.47	60.9%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	40.85s	-24.4%
A→C	22.5s	31.55s	+40.2%
A→D	19.42s	23.45s	+20.8%
B→D	39.44s	47.64s	+20.8%
C→D	46.14s	55.74s	+20.8%
D→E	22.5s	36.21s	+60.9%
E→B	0s	1.57s	+0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	1850	1530	1850	54s / 40.85s	Congested	Congested
A→C	1490	50	3	0.95	Arterial	1750	1267	1750	22.5s / 31.55s	Congested	Congested
A→D	2200	60	4	1.6	Expressway	1900	2090	1900	19.42s / 23.45s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	2500	1530	2500	39.44s / 47.64s	Congested	Congested
C→D	1300	40	2	0.8	Arterial	1950	1105	1950	46.14s / 55.74s	Congested	Congested
D→E	2200	60	4	1.4	Expressway	2150	2090	2090	22.5s / 36.21s	Congested	Normal
E→B	1800	50	3	1	Arterial	0	1530	60	0s / 1.57s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	5	4
Average congestion (veh/hr)	1728.57	1728.57
Maximum congestion (veh/hr)	2500.00	2500.00
Total network congestion	12100.00	12100.00
Mean congestion-to-threshold ratio	1.13	1.13
Average signal cycle time (s)	—	66.64
Fitness value	11.04	10.98
PSO iterations to convergence	—	100

Scenario 7 — Late Evening (8:30–10:30 PM)

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	1300
A→C	1490	1100
A→D	2200	1700
B→D	1800	01650
C→D	1300	1250
D→E	2200	1700
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED Yes	ITERATIONS 100	INITIAL FITNESS 5.24	FINAL FITNESS 5.02
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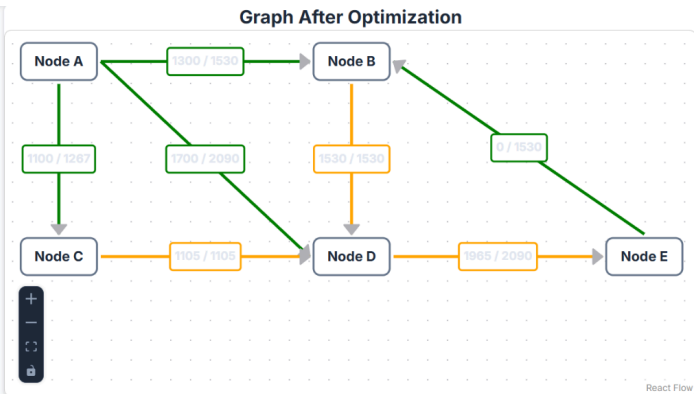
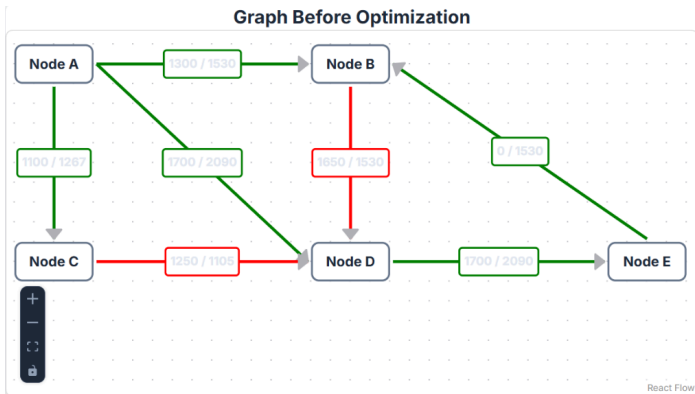
Overall Fitness Improvement: 4.20%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	92.07	130.2%
B	60	72.48	20.8%
C	45	73.08	62.4%
D	70	50.65	-27.6%
E	50	105.76	111.5%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	65.23s	+20.8%
A→C	22.5s	36.54s	+62.4%
A→D	24.99s	19.51s	-21.9%
B→D	37.45s	27.1s	-27.6%
C→D	42.56s	29.36s	-31.0%
D→E	22.5s	47.59s	+111.5%
E→B	0s	0s	0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	1300	1530	1300	54s / 65.23s	Normal	Normal
A→C	1490	50	3	0.95	Arterial	1100	1267	1100	22.5s / 36.54s	Normal	Normal
A→D	2200	60	4	1.6	Expressway	1700	2090	1700	24.99s / 19.51s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	1650	1530	1530	37.45s / 27.1s	Congested	Normal
C→D	1300	40	2	0.8	Arterial	1250	1105	1105	42.56s / 29.36s	Congested	Normal
D→E	2200	60	4	1.4	Expressway	1700	2090	1965	22.5s / 47.59s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	2	0
Average congestion (veh/hr)	1242.86	1242.86
Maximum congestion (veh/hr)	1700.00	1965.00
Total network congestion	8700.00	8700.00
Mean congestion-to-threshold ratio	0.79	0.78
Average signal cycle time (s)	—	78.81
Fitness value	5.24	5.02
PSO iterations to convergence	—	100

Scenario 8 — Night (10:30 PM–12:30 AM)

Novel PSO Traffic Optimization

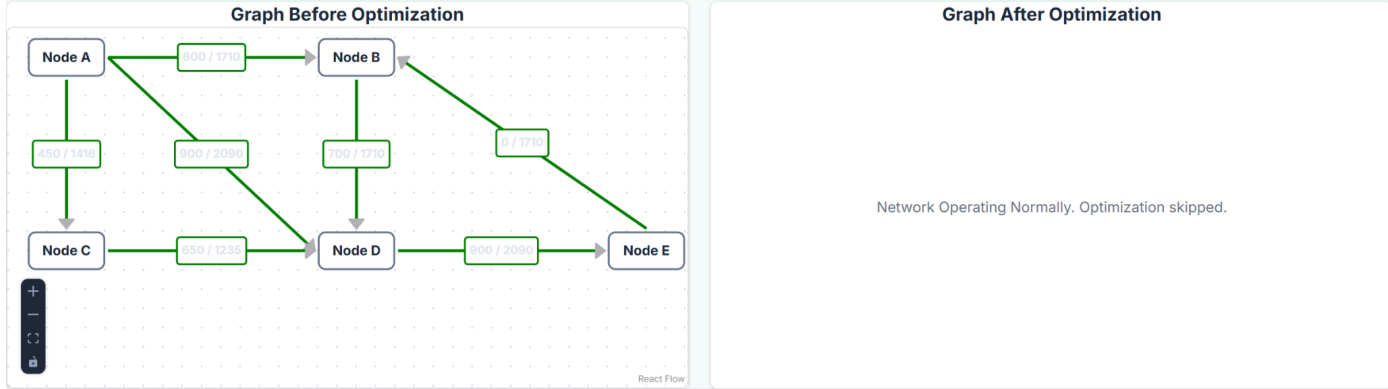
Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	600
A→C	1490	450
A→D	2200	900
B→D	1800	700
C→D	1300	650
D→E	2200	900
E→B	1800	0

Run Optimization

Optimization Status: No Optimization Required. All edges are operating below their capacity thresholds.



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	600	1710	600	Normal	Normal
A→C	1490	50	3	0.95	Arterial	450	1416	450	Normal	Normal
A→D	2200	60	4	1.6	Expressway	900	2090	900	Normal	Normal
B→D	1800	50	3	1.1	Arterial	700	1710	700	Normal	Normal
C→D	1300	40	2	0.8	Arterial	650	1235	650	Normal	Normal
D→E	2200	60	4	1.4	Expressway	900	2090	900	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1710	0	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	0	—
Average congestion (veh/hr)	600.00	—
Maximum congestion (veh/hr)	900.00	—
Total network congestion	4200.00	—
Mean congestion-to-threshold ratio	0.35	—
Average signal cycle time (s)	—	—
Fitness value	0.00	0.00
PSO iterations to convergence	—	0

Scenario 9 — Festival / Heavy Rain (Stress Test)

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	2400
A→C	1490	2300
A→D	2200	2600
B→D	1800	3200
C→D	1300	2700
D→E	2200	3000
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED Yes	ITERATIONS 100	INITIAL FITNESS 19.71	FINAL FITNESS 19
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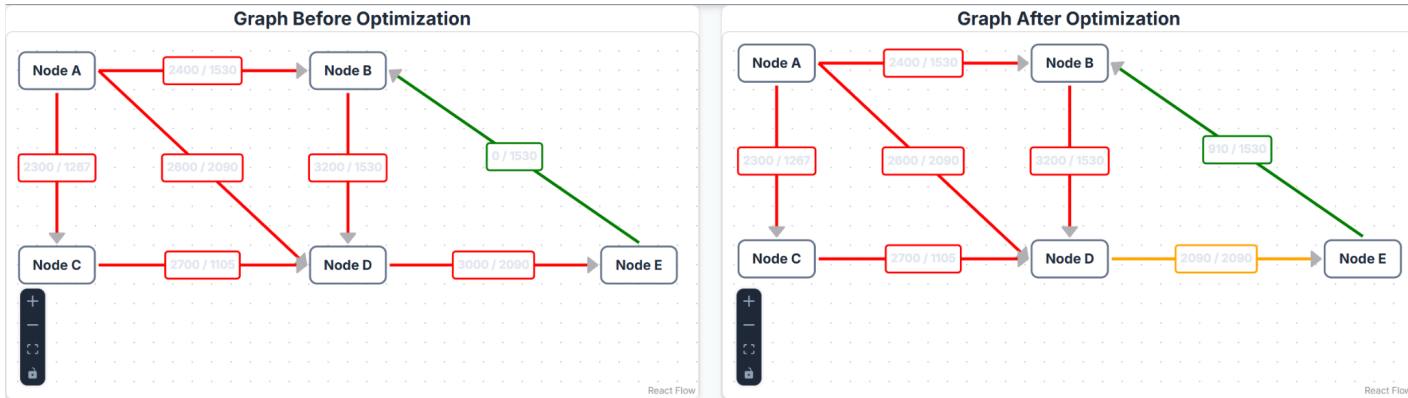
Overall Fitness Improvement: 3.60%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	112.89	182.2%
B	60	96.49	60.8%
C	45	116.73	159.4%
D	70	66.4	-5.1%
E	50	69.29	38.6%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	76.96s	+42.5%
A→C	22.5s	58.36s	+159.4%
A→D	19.79s	18.77s	-5.2%
B→D	37.61s	35.67s	-5.2%
C→D	47.6s	45.15s	-5.1%
D→E	22.5s	31.18s	+38.6%
E→B	0s	29.18s	+0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	2400	1530	2400	54s / 76.96s	Congested	Congested
A→C	1490	50	3	0.95	Arterial	2300	1267	2300	22.5s / 58.36s	Congested	Congested
A→D	2200	60	4	1.6	Expressway	2600	2090	2600	19.79s / 18.77s	Congested	Congested
B→D	1800	50	3	1.1	Arterial	3200	1530	3200	37.61s / 35.67s	Congested	Congested
C→D	1300	40	2	0.8	Arterial	2700	1105	2700	47.6s / 45.15s	Congested	Congested
D→E	2200	60	4	1.4	Expressway	3000	2090	2090	22.5s / 31.18s	Congested	Normal
E→B	1800	50	3	1	Arterial	0	1530	910	0s / 29.18s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	6	5
Average congestion (veh/hr)	2314.29	2314.29
Maximum congestion (veh/hr)	3200.00	3200.00
Total network congestion	16200.00	16200.00
Mean congestion-to-threshold ratio	1.51	1.54
Average signal cycle time (s)	—	92.36
Fitness value	19.71	19.00
PSO iterations to convergence	—	100

Scenario 10 — Road Closure / Accident

Novel PSO Traffic Optimization

Research Prototype

Network Configuration

EDGE	ROAD CAPACITY (VEH/HR)	PREDICTED TRAFFIC (VEH/HR)
A→B	1800	1800
A→C	1490	1700
A→D	2200	1500
B→D	1800	2800
C→D	1300	2000
D→E	2200	1900
E→B	1800	0

Run Optimization

Optimization Summary

TRIGGERED Yes	ITERATIONS 100	INITIAL FITNESS 11.15	FINAL FITNESS 10.81
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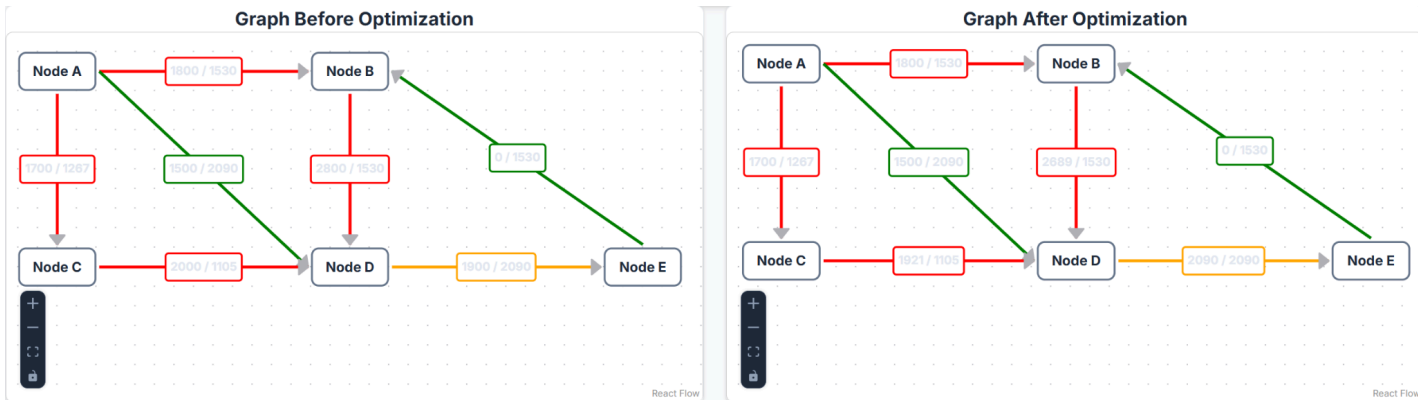
Overall Fitness Improvement: 3.05%

Cycle Times (seconds)

NODE	OLD	NEW	% IMPROVEMENT
A	40	94.04	135.1%
B	60	113.35	88.9%
C	45	52.91	17.6%
D	70	46.41	-33.7%
E	50	46.67	-6.7%

Dynamic Green Times (seconds)

EDGE (APPROACH)	OLD GREEN TIME	NEW GREEN TIME	% CHANGE
A→B	54s	102.02s	+88.9%
A→C	22.5s	26.46s	+17.6%
A→D	15.07s	10.34s	-31.4%
B→D	43.42s	28.61s	-34.1%
C→D	46.52s	30.67s	-34.1%
D→E	22.5s	21s	-6.7%
E→B	0s	0s	0.0%



Traffic Congestion Table

EDGE	CAPACITY	SPEED	LANES	LENGTH (KM)	ROAD TYPE	PREDICTED	THRESHOLD	OPTIMIZED	GREEN TIME (OLD/NEW)	STATUS BEFORE	STATUS AFTER
A→B	1800	50	3	1.2	Arterial	1800	1530	1800	54s / 102.02s	Congested	Congested
A→C	1490	50	3	0.95	Arterial	1700	1267	1700	22.5s / 26.46s	Congested	Congested
A→D	2200	60	4	1.6	Expressway	1500	2090	1500	15.07s / 10.34s	Normal	Normal
B→D	1800	50	3	1.1	Arterial	2800	1530	2689	43.42s / 28.61s	Congested	Congested
C→D	1300	40	2	0.8	Arterial	2000	1105	1921	46.52s / 30.67s	Congested	Congested
D→E	2200	60	4	1.4	Expressway	1900	2090	2090	22.5s / 21s	Normal	Normal
E→B	1800	50	3	1	Arterial	0	1530	0	0s / 0s	Normal	Normal

Research Evaluation Metrics

METRIC	BEFORE PSO	AFTER PSO
Number of congested edges	4	4
Average congestion (veh/hr)	1671.43	1671.43
Maximum congestion (veh/hr)	2800.00	2688.55
Total network congestion	11700.00	11700.00
Mean congestion-to-threshold ratio	1.11	1.10
Average signal cycle time (s)	—	70.68
Fitness value	11.15	10.81
PSO iterations to convergence	—	100